Chapter 1

Introduction

Mathematical models serve an essential role when analyzing scientific phenomena, investigating future or rare events, and simulating potential designs or technologies. They can also be employed to make predictions for regimes in which data may not be available. To provide these capabilities, models must incorporate theory and mechanisms – such as conservation laws or constitutive relations – associated with modeled phenomena. One must also perform code verification to establish the accuracy of potentially large-scale simulation codes. Furthermore, the models must be calibrated and validated using experimental data associated with the process, or numerical data from a validated high-fidelity model. The final requirement for state-of-the-art predictive science[1] entails uncertainty quantification (UQ). This can be broadly defined as the synthesis of mathematical, statistical, and computational theory and methods to systematically identify, quantify, and reduce uncertainties and errors associated with models and their parameters, simulation codes, experimental data, and predicted responses or outcomes. An additional requirement is that this can be achieved for mechanistic models whose complexity prohibits solely sampling-based analysis. The development of a comprehensive uncertainty quantification framework, and illustration for a broad range of physical, engineering, and biological applications, constitutes the subject of this book.

1.1 Motivating Examples and Applications

The examples in this section motivate topics which must be addressed when identifying and quantifying uncertainties for various applications. The first illustrates issues in the context of a simple harmonic oscillator model whereas the second SIR model motivates the analysis in the context of a disease model. We detail these models in Chapter 3 and employ them throughout the text. The third example is

[1]One can broadly define predictive science as the use of verified, calibrated, and validated simulation models to predict phenomena of interest with quantified uncertainties.

historical and illustrates a classical perspective of mechanistic and empirical models, uncertainty quantification, and their combined role for predictive science. Finally, we briefly illustrate in the fourth example the role of uncertainty bounds in weather prediction.

We illustrate in Chapter 2 the central role of uncertainty quantification for nuclear power plant design, quantitative systems pharmacology, and weather and climate predictions. We also detail the manner in which uncertainty quantification is required to construct digital twins for engineering and biomedical systems and virtual populations to facilitate drug discovery and the development of advanced medical treatment regiments.

Example 1.1. The spring model

$$\begin{aligned} &m\frac{d^2z}{dt^2} + c\frac{dz}{dt} + kz = f_0\cos(\omega_F t), \\ &z(0) = z_0 \; , \; \frac{dz}{dt}(0) = z_1, \end{aligned} \tag{1.1}$$

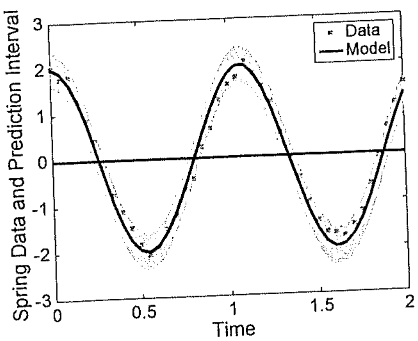
provides a prototype for modeling the displacement $z(t)$ of various vibrating structures. As detailed in Example 3.4, m, c, and k denote the mass, damping, and stiffness of the spring, and f_0 and ω_F are the magnitude and frequency of the driving force. The model parameters, or inputs, are $\boldsymbol{\theta} = [m, c, k, f_0, \omega_F, z_0, z_1]^T$. The following issues motivate concepts pertaining to sensitivity analysis (SA) and uncertainty quantification (UQ), which are typical for many engineering models, and are addressed in this text.

Issues Pertaining to Sensitivity Analysis and Uncertainty Quantification

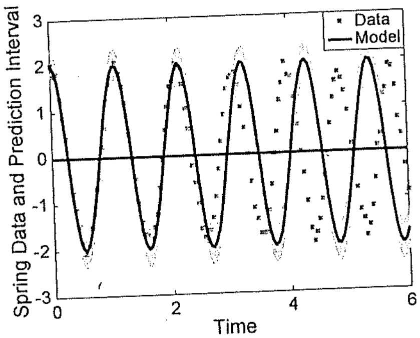
1. The parameters m, c, and k are not uniquely determined by displacement data since one can reparameterize the model with $C = \frac{c}{m}$, $K = \frac{k}{m}$, and $F_0 = \frac{f_0}{m}$. Furthermore, we illustrate in Example 7.3 that one cannot estimate the damping c or initial conditions z_0 and z_1 using displacement data $z_p(t)$ which is periodic in time.

2. The states $z(t)$ and $\frac{dz}{dt}$ can be respectively measured using a proximity sensor and laser vibrometer. Both are accurate but neither provides exact measurements. This yields experimental measurements $Y_i = z(t_i, \boldsymbol{\theta}) + \varepsilon_i$ or $Y_i = \frac{dz}{dt}(t_i, \boldsymbol{\theta}) + \varepsilon_i$, where ε_i are measurement errors. We typically treat ε_i as random variables, which yields random measurements Y_i.

3. The random nature of observed data Y_i introduces uncertainties in model parameters $\boldsymbol{\theta}$, when they are inferred via a fit to data. The determination of distributions for $\boldsymbol{\theta}$, as well as ε_i, constitutes an initial step in uncertainty quantification. The statistical variability in parameters subsequently introduces uncertainty in the model response, as illustrated by the prediction intervals plotted in Figure 1.1(a).



(a)



(b)

Figure 1.1. (a) *Synthetic data, model response, and prediction intervals for the spring model* (1.1). (b) *Model discrepancy due to missing physics.*

4. The model (1.1) is also an approximation and will generally omit some of the physics associated with applications. This can produce model discrepancy, which reduces its accuracy when predicting future events; e.g., see Figure 1.1(b).

5. More complex structures require partial differential equation models to incorporate spatial or other dependencies. For example, certain vibrating structures can be modeled by the Euler–Bernoulli beam model

$$\rho(x)\frac{\partial^2 w}{\partial t^2} + \gamma\frac{\partial w}{\partial t} + \frac{\partial^2}{\partial x^2}\left[YI(x)\frac{\partial^2 w}{\partial x^2} + CI(x)\frac{\partial^3 w}{\partial x^2 \partial t}\right] = f(x,t) \tag{1.2}$$

with appropriate boundary and initial conditions. Here $\rho(x)$, γ, $YI(x)$, and $cI(x)$ denote density, air damping, stiffness, and internal damping terms. The complexity of approximating solutions can inhibit direct uncertainty analysis and requires the intermediate development of surrogate or reduced-order models. The construction of reduced-order models is even more critical for 2-D plate and shell models and PDE models for coupled systems.

Example 1.2. We employ the SIR model

$$\begin{aligned} \frac{dS}{dt} &= \mu(N-S) - \eta kIS \quad &, S(0) = S_0, \\ \frac{dI}{dt} &= \eta kIS - (\gamma+\mu)I \quad &, I(0) = I_0, \\ \frac{dR}{dt} &= \gamma I - \mu R \quad &, R(0) = R_0, \end{aligned} \tag{1.3}$$

detailed in Example 3.6, to illustrate the spread of disease among susceptible $S(t)$, infectious $I(t)$, and recovered $R(t)$ individuals in a population of size N. Here

μ, η, k, and γ respectively denote the birth and death rate, infection coefficient, interaction rate, and recovery rate. The following issues motivate topics which are addressed in subsequent chapters.

Issues Pertaining to Sensitivity Analysis and Uncertainty Quantification

1. Not all of the parameters $\boldsymbol{\theta} = [\mu, \eta, k, \gamma]^T$ can be uniquely determined from measured responses since ηk appears as a product.

2. There can be significant variability and ambiguity when quantifying when an individual is susceptible, infectious, or recovered. As we note in Example 3.6 for flu data, there also can be variability in N depending on assumed demographics. In combination, this produces errors, or uncertainties, which we generally model as random to facilitate analysis. We also discuss in Example 3.6 the interpretation of decimal values of individuals.

3. The uncertainty in diagnosed individuals produces uncertainties when this data is used to infer model parameters. The propagation of parameter uncertainties through models subsequently produces uncertainties in modeled values for $S(t)$, $I(T)$, and $R(t)$, as illustrated by the prediction intervals in Figure 1.2(a).

4. We illustrate in Figure 1.2(b)–(d) the errors which arise when the SIR model is used to quantify disease dynamics in a population which also has exposed individuals $E(t)$. This model discrepancy must be adequately addressed to accurately predict the future spread of disease.

5. The SIR model (1.3) is based on conservation laws in the form of compartment analysis. This ensures that $\frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} = 0$ so that $S(t) + I(t) + R(t) = N$ and the number in each class of individuals remains nonnegative. This is in contrast to data-driven approaches, such as polynomial fitting, neural networks, or Gaussian processes, which do not enforce biological constraints and can yield negative future population demographics.

6. For a spatially varying population, one can consider $S(\boldsymbol{x}, t)$ and $I(\boldsymbol{x}, t)$ where $\boldsymbol{x} = [x_1, x_2]^T$. In the absence of births and deaths, and with a combined infection rate $\beta(\boldsymbol{x})$, one can consider the PDE model

$$
\begin{aligned}
\frac{\partial S}{\partial t}(\boldsymbol{x}, t) &= -\beta(\boldsymbol{x}) I(\boldsymbol{x}, t) S(\boldsymbol{x}, t) + D_s \nabla^2 S(\boldsymbol{x}, t), \\
\frac{\partial I}{\partial t}(\boldsymbol{x}, t) &= \beta(\boldsymbol{x}) I(\boldsymbol{x}, t) S(\boldsymbol{x}, t) - \gamma I(\boldsymbol{x}, t) + D_I \nabla^2 I(\boldsymbol{x}, t).
\end{aligned}
\tag{1.4}
$$

   Here $\nabla^2 \equiv \frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2}$ denotes the Laplacian operator and D_s and D_I are diffusion coefficients. The computational cost of numerically solving (1.4) can inhibit its direct use for sampling-based sensitivity or uncertainty analysis. This may necessitate the intermediate construction of reduced-order or surrogate models, which retain the predictive capabilities of (1.4) but provide the efficiency required for sampling-based analysis.

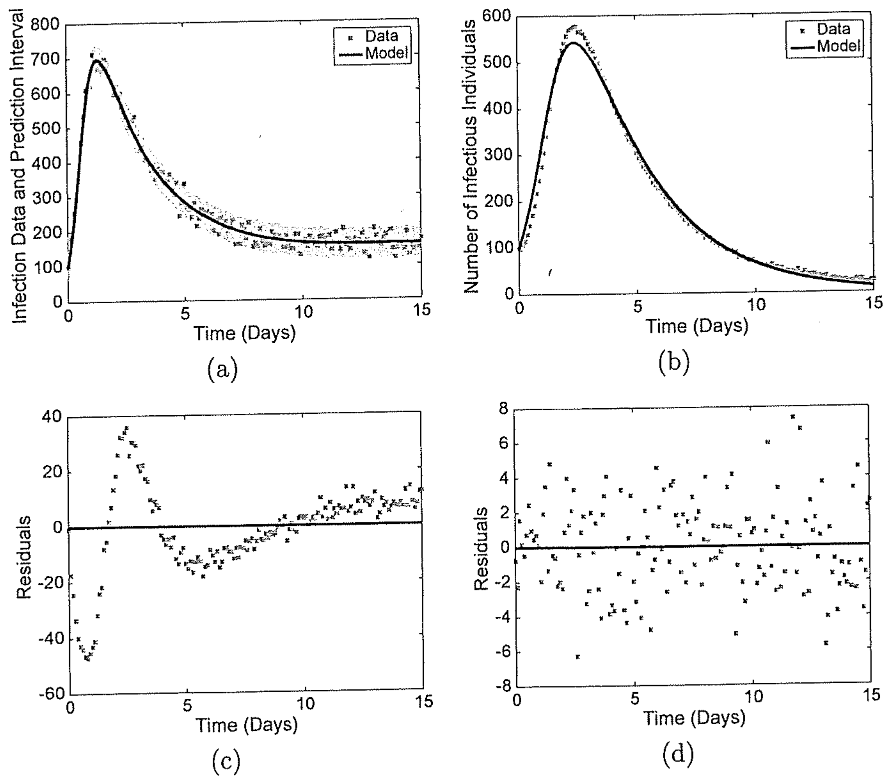


Figure 1.2. (a) *Synthetic data, model response, and prediction intervals for the SIR model* (1.3). (b) *Model discrepancy due to the omission of exposed individuals* $E(t)$. (c) *Residuals for the SIR model which omits* $E(t)$ *and* (d) *includes* $E(t)$.

Example 1.3 (Historical). In 1619, Johannes Kepler completed his laws of planetary motion, based on extensive data collected by Tycho Brahe. Sir Isaac Newton subsequently published his *Philosophiæ Naturalis Principia Mathematica* in 1687, which included the three laws of mechanics in addition to the universal law of gravitation. This theory explained Kepler's laws but, because it was based on fundamental laws of physics, it more broadly quantified the motion of any object in response to an applied force.

When establishing the predictive capabilities of models, one often delineates between mechanistic models, which are based on fundamental physical or biological laws or theory, and empirical or phenomenological models constructed solely using data. The two categories are respectively illustrated by Newton's theory and Kepler's laws. Whereas both are critical for predictive science, fundamental mechanistic models must be employed for extrapolatory or out-of-data analysis, whereas empirical models are restricted to regimes comprising the data used to construct the models.

To illustrate sources of errors or uncertainty, we note that whereas Brahe's data was regarded as among the best available at the time, it was collected without the aid of a telescope, which had not yet been invented. This introduced errors and uncertainties in the data, which subsequently produced uncertainty in parameters, such as semimajor axes and eccentricity, in Kepler's laws. Furthermore, the theory of relativity later provided extensions to Newton's laws for certain physical regimes. Hence, while this seminal work provided foundations for modern theory, it also exhibited uncertainties and errors analogous to those in present applications.

Example 1.4 (Weather Prediction). To illustrate uncertainty intervals in a setting which may be familiar to many readers, we plot in Figure 1.3 the forecast path and cone of uncertainty for the post-tropical storm Debby. These uncertainty bounds were computed using previous data collected under similar atmospheric conditions.

We detail in Section 2.3 the equations of atmospheric physics and a representative constitutive relation for one of the source terms. We note there that both the phenomenological source term and associated parameters contribute uncertainties to weather predictions. These are compounded by the large numerical grids required for real-time weather forecasting, in addition to uncertainties in measured data. Readers can also find details regarding the use of ensemble forecasts, which quantify uncertainties in storm paths based on the propagation of samples from input distributions through the models. This is extended to climate models in Section 2.4.

These examples motivate the following definitions.

Definition 1.5 (Model Parameters or Inputs). We employ the terms parameters or inputs to collectively denote model coefficients, initial conditions, boundary conditions, or exogenous forces, which are considered during sensitivity and uncertainty analysis. *Calibration parameters* are those which are inferred using

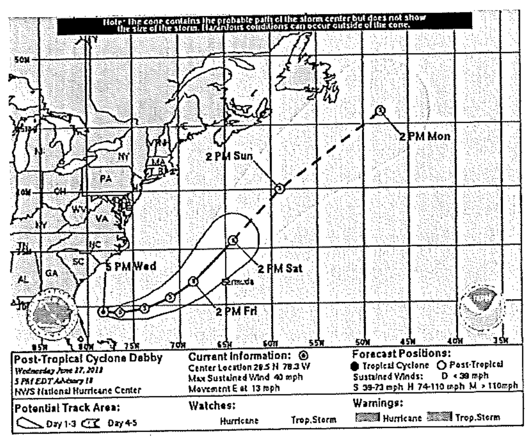


Figure 1.3. *NOAA image of the forecast path and cone of uncertainty for the post-tropical cyclone Debby. Image courtesy of the National Oceanic and Atmospheric Administration.*

experimental data. The terms generally exclude independent variables and control signals.

Definition 1.6 (Model Responses). We denote the states of a model by model responses or model outputs. For an ODE model, states constitute the dependent variables which require initial or final time conditions to ensure a well-posed system. For the spring model (1.1), the state or responses are the displacement $z(t)$ and velocity $\frac{dz}{dt}$, whereas $S(t)$, $I(t)$, and $R(t)$ constitute the responses for the SIR model (1.3).

Definition 1.7 (Quantity of Interest (QoI)). The quantity of interest (QoI) designates the response required to analyze the model or employ it to make decisions or conclusions regarding the process. When employing the QoI for model calibration or validation, it is necessary to have corresponding experimental data or data from a validated high-fidelity model. For many applications, one considers statistical QoI to accommodate uncertainties intrinsic to the modeled process. To illustrate for the SIR model (1.3), one could employ the average infectious numbers $\overline{I}(t)$ as a QoI when making policy decisions regarding a disease.

Definition 1.8 (Verification). Verification refers to the process of quantifying the accuracy of simulation codes used to implement mathematical models. Techniques include, but are not limited to, simulation using independent numerical algorithms – e.g., compare results obtained using finite-difference and finite-element algorithms – and comparison with exact solutions via the Method of Manufactured Solutions [275].

Definition 1.9 (Validation). Validation describes the process of determining the accuracy of mathematical models for quantifying the considered physical or biological processes. This generally involves both the simulation code used to implement the model and experimental data from the process. One can also validate low-fidelity codes using numerical data generated by validated high-fidelity codes. We illustrate the use of prediction intervals for model responses to perform validation in Chapter 13 and refer readers to [275] for a more general discussion of validation techniques.

1.2 Sources and Types of Uncertainties and Errors

We detail in this section the sources and nature of uncertainties motivated by Examples 1.1–1.3. For each category, we provide additional examples based on applications discussed in later chapters.

1.2.1 Uncertainties and Limitations of Experimental Measurements

A theory is something nobody believes, except the person who made it. An experiment is something everybody believes, except the person who made it. Albert Einstein

Fundamental sources of uncertainties and errors in experiments include limited or incomplete data, limited accuracy or resolution of sensors, and data prescribed by physical observation models. The following examples illustrate these sources for a variety of applications.

- The population demographics, for the flu data noted in Example 1.2 and cited in Example 3.6, differ between two sources. Furthermore, there is variability in the designation of what constitutes infectious.

- Pharmaceutical and disease treatment strategies are often too dangerous or expensive for human tests or large segments of the population. Furthermore, various model responses for quantitative systems pharmacology (QSP) models, such as the brain model discussed in Example 2.1, may be difficult or impossible to measure.

- The harsh radioactive, thermal, and chemical environments in a nuclear reactor core limit the availability of measurements for performance improvement, nondestructive evaluation, and safety regulation—see Section 2.1.

- The tolerances of sensors, such as the proximity sensors and laser vibrometer noted in Example 1.1, are often specified by manufacturers. However, these tolerances often depend on the specific configuration and operating regiment, thus producing uncertainties.

- Various sensors employ physical observation models to convert measured quantities to physical units. As discussed in Example 6.3, the speed of an aircraft is determined by applying Bernoulli's principle to relate pressure and velocity for a pitot tube. Other sensors employ nonlinear relations to convert current, voltage, or power to physical quantities. This can introduce varying degrees of uncertainty in the observation process.

- For many applications, low-pass filters are employed to remove high-frequency dynamics. This data processing can introduce errors and uncertainties in subsequent analysis.

Whereas some of these examples illustrate limited, rather than statistically uncertain, data, the analysis is often facilitated by considering experimental observation errors in a statistical framework. This subsequently produces parameter uncertainty during model calibration.

1.2.2 Parameter Uncertainties

Physical and biological models generally have coefficients, which must be specified before the model can be employed to represent or predict the process. Furthermore, one must specify initial or boundary conditions for differential equation models. The following examples illustrate the manner in which model parameters or inputs exhibit uncertainties.

- The damping parameter c in the spring model (1.1) and infection coefficient η in the SIR model (1.3) are difficult to measure directly and must be inferred through a fit to experimental data. This introduces uncertainty due to errors and uncertainties in the data.

- Measured parameters, such as the birth/death rate μ in (1.3), are often reported within an interval to reflect uncertainty or variability in measurements.

- We illustrate in Section 2.1.2 that models for clad chemistry in a nuclear reactor yield parameters, such as the saturation temperature T_{sat}, whose values and certainty must be inferred through a fit to data.

- The minimal physiologically-based pharamacokinetic (m-PBPK) model, discussed in Example 2.1, is specified with nominal values for 37 model parameters. However, it is necessary to additionally determine bounds or distributions for the parameters to quantify the certainty of responses or construct virtual populations in the manner discussed in Section 2.6.

The process of estimating parameter values based on measured data is often termed *model calibration*, *parameter estimation*, or *parameter inference*. The estimation of parameter uncertainties using experimental data is often referred to as *inverse uncertainty quantification*.

1.2.3 Model Errors

Essentially, all models are wrong, but some are useful. George E. P. Box, page 424 of [57].

Model errors result from assumptions and approximations made when quantifying physical, biological, or observation processes. We illustrated one ramification of model errors, also termed *model discrepancy*, in Examples 1.1 and 1.2, where we showed that they can produce incorrect predictions at future times. The following examples illustrate sources and ramifications of model errors.

- Many models are constructed by combining conservation relations – such as compartmental analysis or conservation relations – with constitutive or closure relations. We illustrate in Example 14.1 that the discrepancy in Example 1.1 resulted from the choice of closure relation. The discrepancy in the SIR model (1.3) was due to the choice of states for compartment analysis.

- The thermal-hydraulic and chemistry models, discussed in Section 2.1.3 for nuclear reactor design, are highly complex and involve coupled multiphysics properties that occur on a wide range of spatial and temporal scales. Moreover, several of the processes are represented by phenomenological models having nonphysical parameters. Hence both the models and parameters exhibit uncertainty.

- Many biological applications involve coupled, complex, highly nonlinear processes, which can exhibit delays and are often stochastic and poorly understood. Moreover, they often do not admit an encompassing set of governing relations analogous to those in physical and engineering applications. Hence the associated models can be subject to significant uncertainty.

As we discuss in Chapter 14, the quantification of model errors is typically problem-dependent since it requires that additional knowledge regarding the application be specified. Whereas there exist data-driven techniques to address certain forms of model discrepancy, they are generally restricted to interpolatory or in-data predictions.

1.2.4 Numerical Errors and Uncertainties

We illustrated in Examples 1.1 and 1.2 that the computation of model responses often involves the numerical solution of ODEs and PDEs. This introduces discretization or approximation errors. Additional numerical errors and uncertainties include the following.

- As detailed in Section 2.1.3, thermal-hydraulic codes, such as COBRA-TF, employ lateral meshes having the dimensions of the subchannel. This limits their accuracy for fine-scale resolution and necessitates additional closure relations to quantify phenomena such as rod-to-fluid heat transfer. This also constitutes a form of model discrepancy.
- Bugs or coding errors.
- Bit-flipping, hardware failures, and uncertainty associated with future exascale and quantum computing.

Discretization and approximation errors are typically addressed using numerical analysis. As detailed in [96,168,169,274], however, certain classes of numerical errors can be quantified using the growing field of probabilistic numerics.

1.2.5 Types of Uncertainty

The following definitions are commonly employed to categorize the degree to which uncertainties are inherent in the modeling process or reflect lack of knowledge about mechanisms in the model.

Definition 1.10 (Aleatoric Uncertainty). Aleatoric uncertainty is inherent to a problem or experiment. Hence, in principle, it cannot be reduced by additional physical or experimental knowledge. This is also known as statistical, stochastic, or irreducible uncertainty. Examples include uncertainties associated with phenomenological model parameters, turbulence behavior in nuclear thermal-hydraulics models, and initial conditions for certain quantitative systems pharmacology (QSP) models. Aleatoric uncertainties are often analyzed in a probabilistic framework.

Definition 1.11 (Epistemic Uncertainty). Epistemic or systematic uncertainties are those due to simplifying model assumptions, missing physical or biological mechanisms, or basic lack of knowledge. Many of the previously mentioned modeling errors – e.g., model discrepancy due to limitations in phenomenological expressions for input or closure relations, as discussed in Chapter 14 – and numerical errors are epistemic in nature. Epistemic uncertainties are also associated with parameters which are specified on an interval due to variability or uncertainty. These uncertainties are often more amenable to interval analysis than consideration in a probabilistic framework; e.g., see [49,275] and Chapter 12 of [244].

The distinction between aleatoric and epistemic uncertainties is often not clear since lack of knowledge is relative and may depend on current theory and experimental capabilities. In some cases, one can employ techniques, such as deep learning, to reduce epistemic uncertainties to a form where probabilistic analysis is appropriate. For other applications, such as monitoring risks associated with a nuclear reactor, regulators employ a worst-case analysis based on model responses as parameters are varied in a prescribed interval.

1.3 Sensitivity Analysis and Uncertainty Quantification Process

We employed Examples 1.1 and 1.2 to motivate issues pertaining to sensitivity analysis and uncertainty quantification for engineering and biological models and we summarized general sources of uncertainty in Section 1.2. We detail here the general process of identifying and quantifying uncertainties for predictive models and specify chapters where specific topics are discussed. For this discussion, we employ the observation model

$$Y_i = f(t_i, \theta) + \varepsilon_i \; , \; i = 1, \ldots, n. \tag{1.5}$$

Here f denotes the mathematical model, which is a function of p model parameters $\theta = [\theta_1, \ldots, \theta_p]^T$, and is evaluated at times t_i. Finally, Y_i and ε_i respectively denote random observations and observation errors.

Applications, Models, and Theory from Probability and Statistics

The applications in Chapter 2 illustrate challenges associated with sensitivity and uncertainty analysis and provide examples for later chapters. To facilitate the development and illustration of techniques, we also employ the prototypical models and data detailed in Chapter 3. We summarize in Chapter 4 the relevant theory from probability and statistics, which is employed in subsequent analysis.

Parameter Representation

We illustrate in Chapter 5 the representation of model parameters θ in a framework that accomodates the random nature which results from inference using random data. This includes the construction of finite-dimensional representations for spatially varying parameters such as the density $\rho(x)$ in (1.2) and $\beta(\boldsymbol{x})$ in (1.4).

Observation Models

One rarely has direct observations Y_i of a mathematical model f. To address this, one must construct a statistical observation model which accounts for observation errors, individual variation among subjects or experiments, or indirect measurements of outputs, such as the use of a pitot tube to measure airspeed as noted in Section 1.2.1 and detailed in Example 6.3. We illustrate in Chapter 6 the construction of statistical observation models including the additive relation (1.5). This includes mixed-effects models, which can be employed when modeling population behavior or variability among experiments.

Parameter Selection Techniques

We illustrated in Examples 1.1 and 1.2 cases where the complete parameter vector $\boldsymbol{\theta}$ could not be inferred from observed data due to algebraic dependencies among parameters. Furthermore, the parameter dimensions in applications such as neutronics – e.g., see Section 2.1.3 – can be on the order of $p = 10^3$ to $p = 10^5$, which prohibits subsequent analysis. This motivates the parameter selection techniques in Chapters 7–10.

- **Parameter Identifiability or Influence** (Chapter 7): We detail here mathematical and statistics ways that parameters can fail to influence model responses. The techniques in Chapters 8–10 can then be employed to determine subsets or subspaces of parameters which can be used for subsequent analysis.

- **Local Sensitivity Analysis** (Chapter 8): Local sensitivity analysis quantifies the influence of individual parameters θ_j on the model response f based on the approximation of derivatives or sensitivities $\frac{\partial f}{\partial \theta_j}(\boldsymbol{\theta}^*)$ evaluated at a nominal value $\boldsymbol{\theta}^*$. These approximations also form the basis for parameter subset selection techniques, which incorporate statistical dependencies between parameters.

- **Global Sensitivity Analysis** (Chapter 9): In contrast, global sensitivity analysis establishes the relative influence of parameters $\boldsymbol{\theta}$ when considered over an admissible space Γ. Statistical variance-based methods quantify the manner in which variability in responses is apportioned to variability in parameters whereas derivative-based methods average approximations of $\frac{\partial f}{\partial \theta_j}$ over the admissible space. Both local and global sensitivity analysis techniques can be employed to determine *subsets* of noninfluential parameters which can be fixed for uncertainty analysis.

- **Active Subspace Techniques** (Chapter 10): We illustrate here the use of linear algebra techniques to construct *subspaces* of influential parameters for use in subsequent uncertainty analysis.

Uncertainty Quantification

We illustrated in Examples 1.1 and 1.2, and Section 1.2, that sources of uncertainty include those inherent to experimental data, statistical variability in parameters inferred using that data, model-form errors, errors due to the use of reduced-physics models, and numerical errors resulting from the solution of complex models such as PDEs. We detail here the use of frequentist and Bayesian techniques to statistically infer parameters and observation errors using observation models such as (1.5). This is often referred to as *inverse uncertainty quantification*. We also address the construction of moments and confidence, credible, and prediction intervals for quantities of interest and discuss the effects of model-form errors or model discrepancy.

- **Frequentist Inference** (Chapter 11): Frequentist techniques are based on the assumption that parameters are deterministic but unknown. This framework yields estimates for the mean, covariance matrix, and confidence intervals for parameters. It also provides confidence and prediction intervals for responses.

- **Bayesian Inference** (Chapter 12): In a Bayesian framework, one employs highly efficient sampling algorithms to construct posterior distributions for model parameters and observation errors. This framework has the additional advantage that by specifying informative prior information, one can infer parameters which are nonidentifiable in the sense detailed in Chapter 7. In Chapter 13, we illustrate that one can subsequently sample from posterior distributions to quantify uncertainties associated with model responses.

- **Uncertainty Propagation** (Chapter 13): We provide here a framework for quantifying uncertainties associated with model responses or QoIs. One can establish analytic moment relations for linearly parameterized models and mildly nonlinear problems where a truncated Taylor series is sufficiently accurate. For general models, one can propagate samples from posterior distributions for parameters and observations, constructed in Chapter 12, through models to construct distributions and prediction intervals for QoIs. Because the prediction intervals quantify the probability of future numerical or experimental observations, they provide a framework to statistically validate models. In combination with the numerical surrogate models, such as stochastic Galerkin methods, the techniques in this chapter often constitute the topics considered as *uncertainty quantification* for computational models.

- **Model Discrepancy** (Chapter 14): The quantification of model-form errors, or model discrepancy, is complicated by the fact that techniques used to identify and address the phenomenon are model-dependent. We illustrate the manner in which neglected model discrepancy can degrade the accuracy of models and summarize solutions which are applicable for certain applications.

Surrogate and Reduced-Order Models

We illustrated in Examples 1.1 and 1.2 partial differential equation (PDE) models for beam dynamics and spatially varying disease demographics. The computational complexity of simulating these, and more comprehensive, models often precludes their direct use for sampling-based sensitivity and uncertainty analysis. This is further noted for the applications and simulation models detailed in Chapter 2. To address this, one *first* constructs and verifies surrogate, or reduced-order, models, which incorporate the primary features of the modeled process but provide the efficiency required to implement the techniques of Chapters 6 through 14.

- **General Surrogate Models** (Chapter 15): This chapter motivates and summarizes the general frameworks for constructing numerical and statistical surrogate models. It also details random and quasi-random designs used to sample parameter values for surrogate model construction.

- **Numerical Surrogate Models** (Chapter 16 and 17): We illustrate here surrogate models constructed using polynomial, neural network, and spectral basis functions. The spectral representations are constructed using Hermite and Legendre basis functions, which are respectively orthogonal with respect to Gaussian and uniform distributions. For models having these input distributions, spectral surrogate models provide analytic relations for global sensitivity indices and response moments employed for sensitivity and uncertainty analysis. As detailed in Chapter 17, this can be important for differential equation models whose computation complexity often prohibits sampling-based sensitivity and uncertainty analysis. We summarize statistical and numerical techniques to approximate associated integrals in Chapter 20.

- **Statistical Surrogate Models** (Chapter 18): Gaussian process, or kriging, surrogates provide the advantage that, in addition to providing estimates for the mean model response, they yield prediction intervals which quantify statistical variability in the surrogate. They can also be advantageous for models having high-dimensional input spaces or insufficient structure to permit accurate numerical analysis.

- **Reduced-Order Models** (Chapter 19): The surrogate models detailed in Chapters 16–18 are appropriate for interpolatory predictions within the domain of parameters used to train the model. Because they do not directly incorporate the underlying physical or biological mechanisms, however, they are generally not suitable for extrapolatory or out-of-data predictions. By retaining the full-order model structure, projection-based reduced-order models are generally better suited for extrapolatory predictions, such as forecasting future events using time-dependent PDE models. As with surrogate models, one uses the large-scale model simulations to construct and verify a reduced-order model to be employed for subsequent sensitivity and uncertainty analysis.

1.4 Omitted Topics

Although important for predictive science, there are topics which we omit for the sake of brevity. We assume that simulation codes have been verified in the sense detailed in Definition 1.8. Hence we omit this topic and refer readers to [275] for details regarding code verification. We discuss the manner in which the prediction intervals constructed in Chapter 13 can be used to perform code validation as defined in Definition 1.9. However, we do not discuss other validation techniques and again refer readers to [275] for details on this topic.

We focus primarily on probabilistic and statistical techniques for uncertainty quantification and do not discuss the use of interval analysis when quantifying epistemic uncertainties as discussed in Definition 1.11. This includes, for example, model parameters or numerical errors which are constrained to an interval but do not have an assumed or constructed distribution. Furthermore, representing them by a uniform distribution, which assumes that all points in the interval are equally likely, may not be appropriate. We refer readers to [275] and Chapter 12 of [244] for details on this topic. Techniques to address such uncertainties include horsetail plots, second-order sampling, p-boxes and model evidence, Kolmogorov–Smirnoff confidence bounds, and multiple intervals based on expert judgment.

Multifidelity methods also serve a significant role for sensitivity analysis, inverse problems, and uncertainty quantification. However, a complete treatment would require the discussion of multiple background topics so we instead cite selected references. A broad objective of multifidelity methods is to employ low-fidelity simulations – e.g., using surrogate, reduced-order, or reduced-physics models – in combination with high-fidelity models to provide the computational efficiency required for sensitivity analysis and uncertainty quantification while maintaining accuracy and stability criteria. We refer readers to [284] for a survey of multifidelity methods in the context of parameter inference and uncertainty quantification and [249, 296] for details regarding their use for estimating sensitivity indices. In some fields, these technologies are termed high-to-low methodologies.